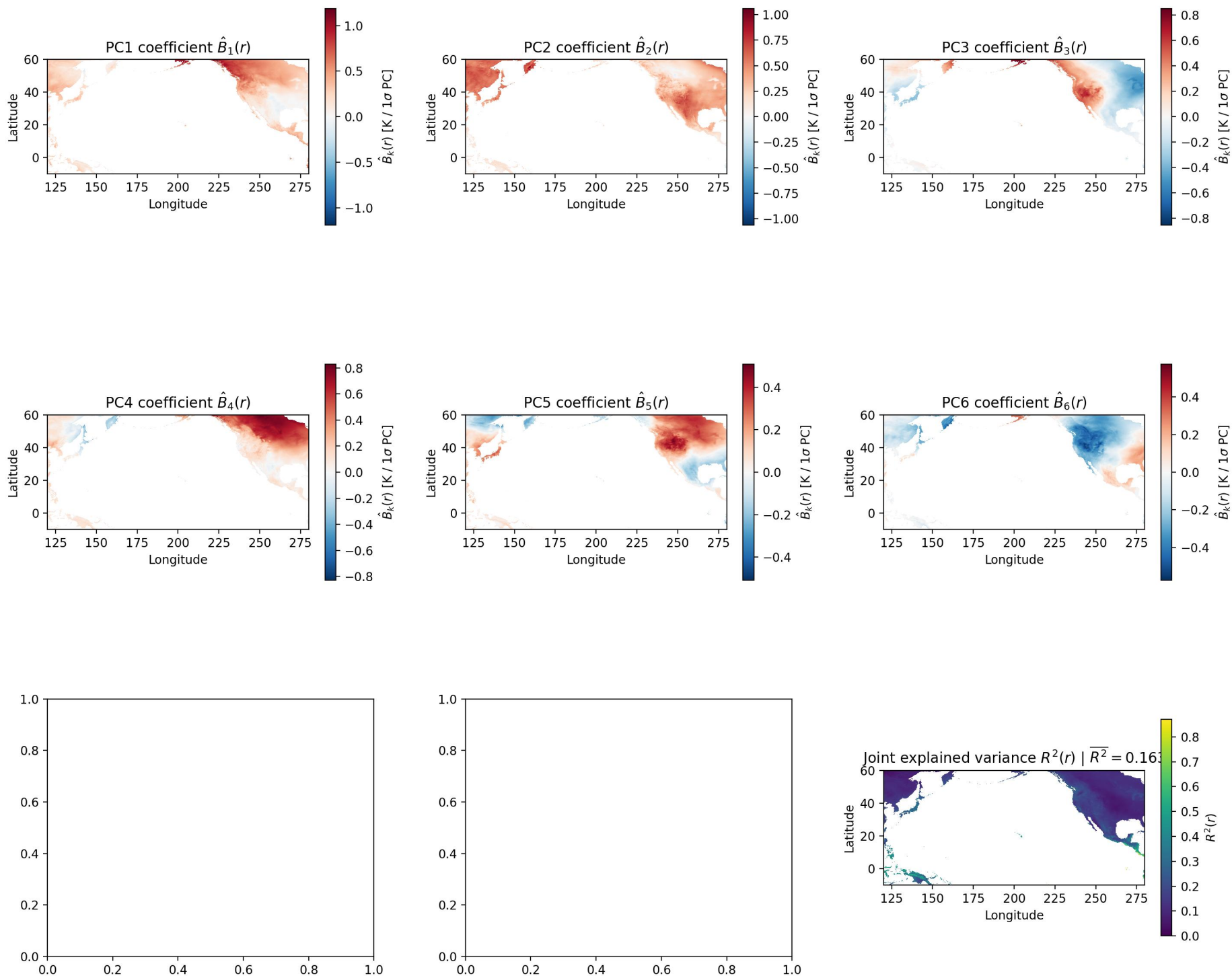




Sensitivity test: joint Level 2 regression with $A = [\text{PC1}, \text{PC2}, \text{PC3}, \text{PC4}, \text{PC5}, \text{PC6}]$.

$$\hat{B} = (A^T A)^{-1} A^T Y \text{ and } \hat{Y} = A \hat{B}.$$

Each $\hat{B}_k(r)$ map is the local matched-region T2m response to a one-unit increase in PC k , after accounting for the other five PCs. PCs are standardized here, so each coefficient is in K per one-standard-deviation increase in the corresponding PC.

$R^2(r) = 1 - \frac{\sum_t [Y(t, r) - \hat{Y}(t, r)]^2}{\sum_t [Y(t, r)]^2}$ gives the fraction of local matched-region T2m anomaly variance explained by PC1-PC6 together.